Score %	IB Grade

EXERCISE 1 **[4 marks]**

Consider the function

$$f(x) = \frac{3x-5}{x+2}, \qquad x \neq -2.$$

(a) Write down the equations of the vertical and horizontal asymptotes of $y = f(x)$.

(b) Hence write down the value of:

a. $\lim_{x \to \infty} f(x)$

b. $\lim_{x \to -2^+} f(x)$

EXERCISE 2 **[5 marks]**

Differentiate $f(x) = 2x^2 - x$ from first principles, showing all steps. Hence find $f'(2)$.

EXERCISE 3 **[4 marks]**

Find $\frac{dy}{dx}$ for each of the following:

1. $y = 5x^3 - 4x^2 + 7$

2. $y = 3\sqrt{x} - \frac{2}{x^2}$

3. $y = (2x - 1)(x + 3)$

EXERCISE 4 **[4 marks]**

Use the product rule to find $\frac{dy}{dx}$ if $y = (2x + 1)(x^2 - 3)$

EXERCISE 5 **[5 marks]**

Use the quotient rule to find $\frac{dy}{dx}$ if:

$$y = \frac{2x - 3}{x + 1}$$

EXERCISE 6 **[3 marks]**

1. Use the chain rule to find $\frac{dy}{dx}$ if:

$y = \frac{1}{(3x-2)^3}$

2. Hence, find the gradient of the tangent at $x = -1$.

Answer Key

1. Asymptotes and limits of $f(x) = \dfrac{3x-5}{x+2}$ *[4 marks]*

a. Vertical asymptote: $x = -2$ A1
Horizontal asymptote: $y = 3$ A1
A0 if the vertical asymptote is left as "$x + 2 = 0$" without stating the equation; A0 if the horizontal asymptote is stated without any justification of the ratio of leading coefficients.

b. $\lim\limits_{x\to\infty} f(x) = 3$ A1
$\lim\limits_{x\to -2^+} f(x) = -\infty$ A1
A0 for "undefined" or "does not exist" in place of $-\infty$; the sign must be correct (numerator $\to -11$, denominator $\to 0^+$).

2. First principles: $f(x) = 2x^2 - x$ *[5 marks]*
$f(x+h) = 2(x+h)^2 - (x+h) = 2x^2 + 4xh + 2h^2 - x - h$ M1
$f(x+h) - f(x) = 4xh + 2h^2 - h = h(4x + 2h - 1)$ M1
$\dfrac{f(x+h)-f(x)}{h} = 4x + 2h - 1 \xrightarrow[h\to 0]{} f'(x) = 4x - 1$ M1A1
$f'(2) = 4(2) - 1 = 7$ A1
The three M marks are for: (i) correctly expanding $f(x+h)$; (ii) correctly forming and factorising $f(x+h) - f(x)$ so that a factor of h is visible; (iii) explicitly taking the limit as $h \to 0$. No marks are awarded for $f'(x) = 4x - 1$ obtained by standard differentiation rules instead of first principles, even if correct — this is not first-principles work.

3. Basic differentiation rules *[4 marks]*

a. $y = 5x^3 - 4x^2 + 7 \;\Rightarrow\; \dfrac{dy}{dx} = 15x^2 - 8x$ A1

b. Rewrite as $y = 3x^{1/2} - 2x^{-2}$ M1
$\dfrac{dy}{dx} = \frac{3}{2}x^{-1/2} + 4x^{-3} = \dfrac{3}{2\sqrt{x}} + \dfrac{4}{x^3}$ A1
M0 if index laws applied incorrectly; A1 requires both *terms correct.*

c. $y = (2x-1)(x+3) = 2x^2 + 5x - 3 \;\Rightarrow\; \dfrac{dy}{dx} = 4x + 5$ A1
A0 for an answer obtained without first expanding the brackets.

4. Product rule: $y = (2x+1)(x^2-3)$ *[4 marks]*
$u = 2x+1,\ u' = 2; \quad v = x^2 - 3,\ v' = 2x$ M1
$\dfrac{dy}{dx} = u'v + uv' = 2(x^2-3) + (2x+1)(2x)$ M1A1
$= 6x^2 + 2x - 6$ A1
First M1 for correctly identifying u, u', v, v'; second M1 for correct substitution into the product-rule formula; first A1

for a correct unsimplified expression; final A1 requires full expansion and collection of like terms. Maximum 3/4 if the final answer is left unsimplified.

5. Quotient rule: $y = \dfrac{2x-3}{x+1}$ *[5 marks]*

$u = 2x-3,\ u' = 2; \quad v = x+1,\ v' = 1$ M1

$\dfrac{dy}{dx} = \dfrac{u'v - uv'}{v^2}$ M1

$= \dfrac{2(x+1) - (2x-3)}{(x+1)^2}$ A1

$= \dfrac{2x+2-2x+3}{(x+1)^2}$ A1

$= \dfrac{5}{(x+1)^2}$ A1

First M1 for correct u, u', v, v'; second M1 for correct quotient-rule structure; first A1 for a correct unsimplified numerator over $(x+1)^2$; second A1 for correctly expanding the numerator; final A1 for the fully simplified answer — do not award the final A1 for an equivalent but unsimplified form.

6. Chain rule: $y = \dfrac{1}{(3x-2)^3} = (3x-2)^{-3}$ *[3 marks]*

a. Let $u = 3x-2,\ u' = 3$.

$\dfrac{dy}{dx} = -3u^{-4} \cdot u' = -9(3x-2)^{-4} = \dfrac{-9}{(3x-2)^4}$ M1A1

M1 for correct use of the chain rule; A1 for the fully correct derivative — no credit if the factor of 3 from u' is omitted or the sign is wrong.

b. At $x = -1$: $3x-2 = -5$, so $(3x-2)^4 = 625$.

$\dfrac{dy}{dx} = \dfrac{-9}{625}$ A1FT

Follow-through from the candidate's part (a); award A1FT only if their expression is correctly evaluated at $x = -1$ (not $x = 1$).